Revisiting semiclassical scalar QED in 1+1 dimensions

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Abstract

We study the backreaction of a charged quantum scalar field in the presence of two opposite charges placed at the boundaries of a finite one-dimensional region, with attention to boundary effects. We review previous work by Ambjørn & Wolfram [1] which finds, in this setup, that vacuum polarization averts instabilities. As it was later pointed out, [1] used a wrong prescription for computing the vacuum polarization [13]. The correct prescription presents notable discrepancies with the one previously used [16]. We find that using the correct prescription the system remains stable despite these discrepancies. To this end, an iterative procedure is introduced to study the system under varying mass and boundary conditions configurations.

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1 Introduction

Vacuum polarization is a phenomenon in quantum electrodynamics (QED) by which the vacuum of a charged field polarizes when subject to external electric fields, effectively behaving like a dielectric with $\epsilon > 0$. This mechanism was one of the first QED effects studied [3], with theoretical consequences that followed shortly after [15, 7]. This mechanism has been already observed, e.g. in corrections to atomic levels [10] (cf. Ref. [11] for a review), while other effects e.g. the Schwinger effect [14] lie beyond current technological advances (cf. Ref. [4]).

Defining vacuum polarization for generic backgrounds[1] is no easy task. Historically, [3] first defined it through the assumption that the short-distance behavior of the physically relevant states in the presence of strong background electric fields is the same as in their absence, only up to smooth coefficients. In modern bibliography, these states are called Hadamard, (cf Ref. [5]), and they are particularly interesting in the context of quantum field theory on curved spacetimes, where the same idea is used to define observables necessary for the study of e.g. perturbative interacting field theory or semiclassical gravity. Detailed discussions on the definition on such observables can be found in Refs. [8, 9].

Semiclassical gravity, a possible first step towards a quantum theory of gravity, couples classical gravity to the expectation value of the stress-energy tensor, which similarly as the vacuum polarization needs to be properly defined through renormalization. Work has been

[1] And any other observable defined through the evaluation of products of fields or their derivatives at coinciding points.

done in this direction e.g. [12, 6], where solutions to the semiclassical Einstein field equations often rely on highly symmetrical spacetimes. Motivated by studying configurations with less symmetric requirements we consider, as a toy model, the effect of the backreaction of a charged scalar field to a background classical electric field in 1+1 dimensional Minkowski spacetime. This setup was already studied in [1], where it was found that neglection of backreaction leads to instabilities, which its inclusion cures. However, as pointed out in [13], the expression used in [1] for vacuum polarization ρ cannot be derived from a manifestly gauge invariant prescription. [16] derives the correct expression for ρ, which has qualitative disagreements with the one used in [1].

This discussion motivates the following question, which we answer in the present paper: Given the difference in the prescriptions for ρ, does using the correct one avoid these instabilities? We will use the correct prescription described in [16] to implement an iterative method to study the effect of the backreaction of the scalar field. It has been found that this expression also alleviates the above mentioned instabilities.

The layout of this article is as follows: Section 2 sets the problem up, presents the semiclassical approximation and introduces a convenient gauge fixing. Sec. 3 discusses the mode decomposition in the fixed gauge, and reviews the definition of vacuum polarization through a gauge invariant renormalization prescription. A comparison with existing bibliography is done. Sec. 4 introduces an iterative procedure for solving the semiclassical KGM equations, and the results are shown in Sec. 5, which are compared with existing results. These results are finally discussed in Sec. 6

This article summarizes and extends the results of the M.Sc. thesis of Santiago Sanz-Wuhl, written in 2025 at the Institut für Theoretische Physik, Universität Leipzig, under the supervision of Jochen Zahn.

2 Set up

In 1+1 dimensional spacetime, two charges $q, -q$ are set at $x^1 = 0, x^1 = a$, respectively. Ignoring vacuum polarization effects, this yields a constant electric field of strength q pointing towards positive x^1 in the region $x^1 \in [0, a]$, 0 everywhere else. We study the solutions to the semiclassical Klein-Gordon-Maxwell equations (KGM), i.e. the system of coupled differential equations

$$\left(D_\mu D^\mu + m^2\right)\phi = 0, \tag{1a}$$

$$\partial_\mu F^{\mu\nu} = \langle j^\nu \rangle\,, \tag{1b}$$

in the region $x = (x^0, x^1) \in \mathbb{R} \times [0, a]$. Here, $D_\mu = \partial_\mu + ieA_\mu$ is the gauge covariant derivative,

$$j^\nu(x) = \frac{ie}{2}\left((D^\nu\phi^*(x))\phi(x) - \phi^*(x)D^\nu\phi(x)\right) \tag{2}$$

is the charge current density of the Klein-Gordon field, and $\langle\mathcal{O}\rangle$ is the renormalized vacuum expectation value of an observable $\mathcal{O}$, and the metric signature is $(+,-)$. We solve for $F_{\mu\nu}$ and (the mode decomposition of) the two-point function of the vacuum state.

In 1+1 dimensions, A_1 can be arbitrarily set to 0. This way, Eq. (1b) reduces to the Poisson equation for the electrostatic potential A_0

$$\partial_1^2 A_0(x^1) = -\left\langle j^0\right\rangle \equiv -\rho, \tag{3}$$

with ρ the vacuum polarization, i.e. the vacuum expectation value of the zeroth component of the charge current density. Eq. (3) will be solved with the boundary conditions

$$\partial_1 A_0(x^1)\big|_{x^1=0,a} = -q. \tag{4}$$

This way, measuring the electric field close enough to the boundary is not affected by vacuum polarization. The boundary conditions (4) fix the solution to Eq. (3) up to an addition constant, which is for convenience chosen so that $A_0(\frac{a}{2}) = 0$. With these boundary conditions and for a given $\rho(x^1)$, Eq. (3) is solved by

$$A_0(x^1) = -q\left(x^1 - \frac{a}{2}\right) - \int_{\frac{a}{2}}^{x^1}\int_0^{\chi}\rho(\xi)d\xi d\chi. \tag{5}$$

3 Field Modes and Vacuum Polarization

In our gauge, the Klein-Gordon equation is of the form

$$\left[(\partial_0 + ieA_0)^2 + \partial_1^2 - m^2\right]\phi = 0, \tag{6}$$

which we will study using Dirichlet boundary conditions (DBC) $\phi|_{x^1=0,a} = 0$ or Neumann boundary conditions (NBC) $\partial_1\phi|_{x^1=0,a} = 0$. General boundary conditions may be considered, though they are outside of the scope of this work. Using the mode ansatz $\phi_n(x) = \phi_n(x^1)e^{-i\Omega_n x^0}$ the PDE is split into ordinary differential equations for each mode $\phi_n(x^1)$ of the scalar field

$$\left[(\Omega_n - eA_0)^2 + \partial_1^2 - m^2\right]\phi_n = 0. \tag{7}$$

The following dimensionless parameters are defined

$$z = a^{-1}x^1 \tag{8a}$$
$$\lambda = eEa^2 \tag{8b}$$
$$\varepsilon = ae \tag{8c}$$
$$\omega_n = a\Omega_n \tag{8d}$$

Eq. (7) then takes the form

$$\left[(\omega_n - \varepsilon A_0)^2 + \frac{d^2}{dz^2} - a^2m^2\right]\phi_n = 0. \tag{9}$$

The modes ϕ_n are normalized with respect to the symplectic norm

$$(\phi_n, \phi_m) = i\int_0^1 (\phi_n^*\pi_m^* - \pi_n\phi_m)dz, \tag{10}$$

with $\pi_n = (D_0\phi_n)^*$ the canonical conjugate of ϕ_n.

The general solution to Eq. (6) is thus decomposed in modes

$$\phi(t,z) = \sum_{n>0} a_n\phi_n(z)e^{-i\omega_n t} + \sum_{n<0} b_n^\dagger\phi_n(z)e^{-i\omega_n t}. \tag{11}$$

Positive (negative) values of n are identified with modes of positive (negative) norm. a_n and b_n are the ladder operators for the positive and negative norm solutions, respectively, obeying canonical commutation relations $[a_n, a_m^\dagger] = [b_n, b_m^\dagger] = \delta_{nm}$, and all other commutators vanishing. The ladder operators define the vacuum state $|0\rangle$ as that for which $a_n|0\rangle = b_n|0\rangle = 0$. Note that due to the chosen gauge, $\omega_n = -\omega_{-n}$, and $b_n^\dagger$ is the creation operator of the n-th negative frequency mode.

3.1 Vacuum Polarization

Quantities such as vacuum polarization ρ need to be properly defined through a gauge invariant renormalization scheme, due to the presence of products of fields at coinciding points. To define such local field products, one assumes that the states of a field ϕ are Hadamard, i.e. they have a similar short-distance behavior to the vacuum two-point function in absence of background fields, and are generally considered to be the only physically relevant states [5].

The two-point function

$$w_\Omega^{\phi\phi^*}(x,x') = \langle\Omega|\phi(x)\phi^*(x')|\Omega\rangle$$

of the scalar field ϕ in a Hadamard state Ω is of the form

$$w_\Omega^{\phi\phi^*}(x,x') \equiv H^{\phi\phi^*}(x,x') + R_\Omega^{\phi\phi^*}(x,x'). \tag{12}$$

Here,

$$H^{\phi\phi^*} = -\frac{V(x,x')}{4\pi}\ln[(t-t')^2 - (z-z')^2 + i\varepsilon\mathrm{sign}(t-t')] \tag{13}$$

is the Hadamard parametrix of the scalar field in 1+1 spacetime dimensions and $R_\Omega^{\phi\phi^*}$ a state-dependent smooth function. The smooth coefficient V, up to order relevant for vacuum polarization reads [13]

$$V(x,x') = \exp\left[-i\varepsilon\int_0^1 A_\mu(x' + s(x-x'))(x-x')^\mu ds\right] + \mathcal{O}\left((x-x')^2\right).$$

The Hadamard condition allows us to define expectation values of local observables non-linear in the fields as a limit of coinciding points:

$$\langle D_\alpha\phi(x)D_\beta^*\phi^*(x)\rangle = \lim_{x'\to x} D_\alpha D_\beta^{*\prime}(w_\Omega^{\phi\phi^*}(x,x') - H^{\phi\phi^*}(x,x')), \tag{14}$$

with α, β symmetrized multiindices.

Eqs. (14), (2) together with the mode decomposition (11) result in the expression for ρ calculated in [16]

$$\rho = \lim_{\tau\to 0}\Big[\sum_{n>0}(\omega_n - \varepsilon A_0)|\phi_n|^2 e^{-i\omega_n\tau} + \sum_{n<0}(\omega_n - \varepsilon A_0)|\phi_n|^2 e^{i\omega_n\tau}\Big] + \frac{\varepsilon^2}{\pi}A_0. \tag{15}$$

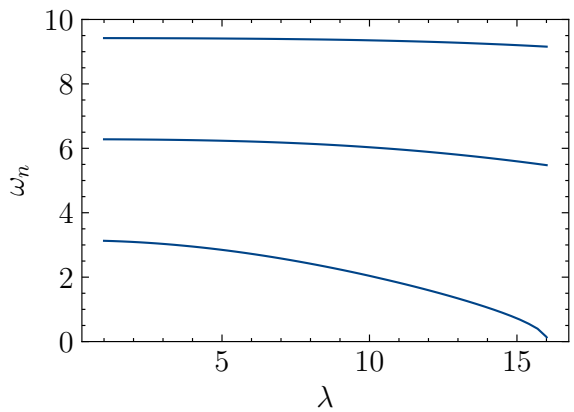

Fig. 1: Energies of the three first modes of the massless Klein-Gordon field subject to DBC in the external field approximation as functions of λ. $\omega_1 \to 0$ as $\lambda \to \lambda_c \approx 16.05$ defines the critical field strength for these mass and boundary conditions configurations.

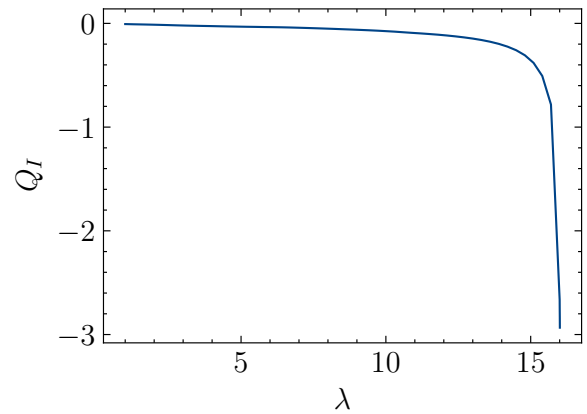

Fig. 2: Corresponding induced charge Q_I in Fig. 1 as a function of λ.

An example of ρ corresponding to the potential $A_0(z) = -\lambda(z - \frac{1}{2})$ with $\lambda = 1$ is shown in Fig. 3.

To measure the *strength* of the vacuum polarization, the *induced charge*

$$Q_I = \int_0^{\frac{1}{2}} \rho(z) dz \tag{16}$$

is defined, which is expected to be negative. This charge partially screens the charge at $z = 0$ and the total electric field at $z = \frac{1}{2}$ is $\lambda + Q_I$.

3.2 External Field Approximation and Critical Field Strength

The external field approximation ignores the contribution of the backreaction of the charged scalar field to the background electric field. This approximation displays the instabilities of the system, which the inclusion of backreaction aims to avoid. Eq. (9) is solved with $A_0(z) = -\lambda(z - \frac{1}{2})$, and takes the form

$$\left[\left(\omega_n + \lambda\left(z - \frac{1}{2}\right)\right)^2 + \frac{d^2}{dz^2} - a^2 m^2\right]\phi_n = 0. \tag{17}$$

For each value of λ, the modes ϕ_n and their energies ω_n can be obtained from the boundary conditions. Fig. 1 displays the energies $\omega_1, \omega_2, \omega_3$ of the first three modes of the massless scalar field with DBC as functions of the background field strength λ. The value at which ω_1 goes to 0 defines the critical field strength $\lambda_c = 16.05039(5)$. Fig. 2 displays the divergent behavior of $Q_I(\lambda)$ for this configuration as $\lambda \to \lambda_c$. λ_c adopts different values for different boundary conditions and masses. [1] shows that for $\lambda > \lambda_c$, modes become unstable and it is no longer possible to define the ground state.

3.3 Mode sum formula

Vacuum polarization in [1] is defined through a so-called mode sum formula. It pairs the charge density associated to each mode as

$$\rho_n^{\mathrm{MS}} = (\omega_n - \varepsilon A_0)|\phi_n|^2 - (\omega_n + \varepsilon A_0)|\phi_{-n}|^2. \tag{18}$$

to define the vacuum polarization

$$\begin{aligned}\rho^{\mathrm{MS}} &= \sum_{n>0} \rho_n^{\mathrm{MS}} \\ &= \sum_{n>0} \left[(\omega_n - \varepsilon A_0)|\phi_n|^2 - (\omega_n + \varepsilon A_0)|\phi_{-n}|^2\right],\end{aligned} \tag{19}$$

where the superscript MS is introduced to differentiate it from the correct prescription ρ. The above expression cannot be derived from a manifestly gauge invariant renormalization scheme and relies on the specific pairing of modes to be well-defined. Furthermore, in the iterative procedure discussed above, [1] truncates the mode sum to $\rho^{\mathrm{MS}} \approx \rho_1^{\mathrm{MS}}$.

Fig. 3 shows different charge densities arising from the different ρ prescriptions, for the massless scalar field subject to DBC. The same ϕ_n, ω_n and $A_0(z) = -(z - \frac{1}{2})$, the external field approximation with $\lambda = 1$, have been used for the different ρ. In blue, the vacuum polarization used in the computations throughout this paper, in orange the vacuum polarization arising from the mode pairing and in red the one used for the iterative procedure in [1]. Notice the non-zero vacuum polarization at the boundaries for Eq. (19), which the reader might find surprising to find in a field subject to DBC. This is an artifact of the missing $\frac{\varepsilon^2}{\pi} A_0$ term in Eq. (19).

We observe that ρ is significantly 'weaker' than the one used by [1], where it was seen that the screening due to ρ^{MS} avoids the instabilities appearing in the external field approximation. This motivates the question answered in this paper: Does considering the correct vacuum polarization ρ also avoid these instabilities?

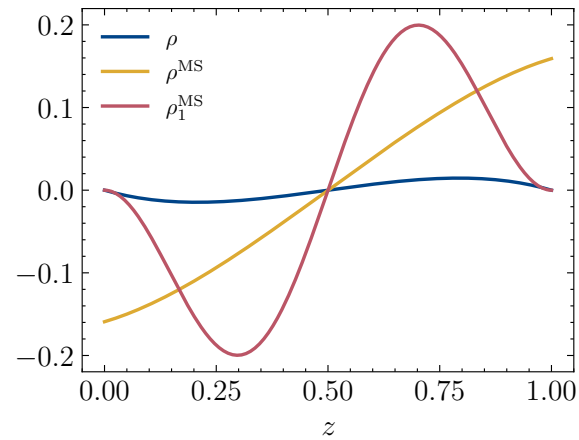

Fig. 3: Three different ρ for the same massless scalar field subject to DBC resulting from $A_0 = -(z - \frac{1}{2})$: In blue, the correct prescription in Eq. (15) used throughout this paper. In orange, ρ calculated through the mode sum (19). In red, ρ defined through the mode sum formula, truncated to the first mode.

4 Self-consistent fields

As discussed above, the polarization of the vacuum induces a charge density ρ, which modifies the background field as in Eq. (5). This modified background field can be used in Eq. (20a) to calculate new solutions, that will have a different vacuum polarization. This iterative procedure, schematized in Fig. 4, is performed *ad infinitum*, and we study its convergence.

More precisely, for each value of λ and at each iteration κ, we compute the following

$$\frac{d^2\phi^\kappa_{n,\lambda}}{dz^2} = -\left[(\omega^\kappa_{n,\lambda} - \varepsilon A^{\kappa-1}_{0,\lambda})^2 - a^2m^2\right]\phi^\kappa_{n,\lambda}, \tag{20a}$$

$$\begin{aligned}\rho^\kappa_\lambda = \lim_{\tau\to 0}\Big[&\sum_{n>0}(\omega^\kappa_{n,\lambda} - \varepsilon A^{\kappa-1}_{0,\lambda})|\phi^\kappa_{n,\lambda}|^2 e^{-i\omega^\kappa_{n,\lambda}\tau} \\ &+ \sum_{n<0}(\omega^\kappa_{n,\lambda} - \varepsilon A^{\kappa-1}_{0,\lambda})|\phi^\kappa_{n,\lambda}|^2 e^{i\omega^\kappa_{n,\lambda}\tau}\Big] \\ &+ \frac{\varepsilon^2}{\pi}A^{\kappa-1}_{0,\lambda},\end{aligned} \tag{20b}$$

$$\begin{aligned}A^\kappa_{0,\lambda} &= cA^{\kappa-1}_{0,\lambda} \\ &\quad + (1-c)\left[-\lambda\left(z - \frac{1}{2}\right) - \int_{\frac{1}{2}}^{z}\int_0^{\chi}\rho^\kappa_\lambda(\xi)d\xi d\chi\right].\end{aligned} \tag{20c}$$

This procedure has two main differences from the one used in [1]: the expression for the vacuum polarization, and the introduction of $c \in [0, 1]$, which will be discussed below.

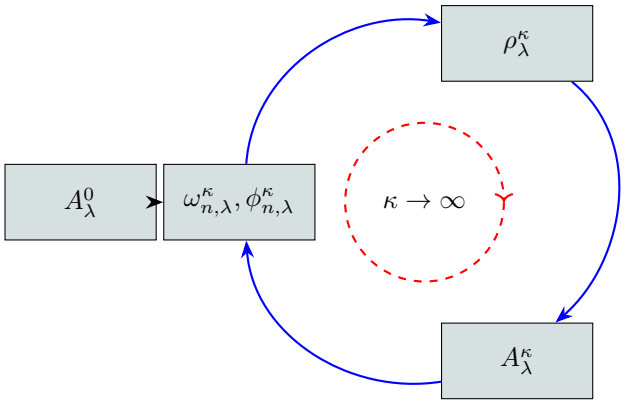

Fig. 4: Schematic of the iterative procedure for fixed λ.

Interpreting Eq. (20c) as an update map on $A_{0,\lambda}^{\kappa-1}$, we identify the iterative procedure with an infinite-dimensional fixed-point problem. That is, for each λ, we seek $A_{0,\lambda}$ such that

$$A_{0,\lambda} = f_{\lambda,c}(A_{0,\lambda}), \tag{21}$$

with $f_{\lambda,c}$ given by the composition of steps (20a), (20b) and (20c). The fixed points of $f_{\lambda,c}$, denominated "self-consistent solutions", are looked for by studying the limit $A_{0,\lambda} = \lim_{\kappa\to\infty} A_{0,\lambda}^\kappa$, where

$$A_{0,\lambda}^{\kappa+1} = f_{\lambda,c}(A_{0,\lambda}^\kappa) = f_{\lambda,c}^\kappa\left(A_{0,\lambda}^0\right), \tag{22}$$

with

$$f_{\lambda,c}^\kappa = \underbrace{f_{\lambda,c} \circ \ldots \circ f_{\lambda,c}}_{\kappa \text{ times}}.$$

By Banach's fixed point theorem [2] the limit $A_{0,\lambda}$ will exist if the map $f_{\lambda,c}$ is a contraction, which is not ensured. If this limit exists, it is given the name self-consistent potential, as well as the corresponding self-consistent vacuum polarization ρ_λ, field modes $\phi_{n,\lambda}$ and mode energies $\omega_{n,\lambda}$.

For low enough $\lambda \ll \lambda_c$, the initial value $A_{0,\lambda}^0$ can be set to the external field approximation $A_{0,\lambda}^0(z) = -\lambda\left(z - \frac{1}{2}\right)$. In the strong-field regime, this is no longer a valid initial condition, due to numerical instabilities appearing for $\lambda > \lambda_c$. This is discussed in the Appendix A. A numerical continuation method is used to traverse the λ-parameter:

1. Assume a self-consistent $A_{0,\lambda}^\infty$ is known.
2. Choose an increment $\Delta\lambda > 0$, which defines a candidate $\lambda_{\text{new}} = \lambda + \Delta\lambda$.
3. Perform the iterative procedure with the initial condition
$$A_{0,\lambda_{\text{new}}}^0 = A_{0,\lambda}^\infty.$$
4. If this sequence converges, store the λ and self-consistent fields, and proceed from step 1 with the newly calculated A_0. If it fails, reduce $\Delta\lambda$, and proceed from step 2.

Finally, in Eq. (20c), $c \in [0, 1]$ is introduced in order to avoid an overshooting of the potential: For $c = 0$, the procedure is precisely the one used in [1], and it is expected to quickly converge. For c close to 1, updates will be small corrections to the previous potential.

5 Results

Using the iterative procedure laid out above we focus on two particular cases: The massless scalar field subject to DBC, and $m = 1$ with the scalar field subject to NBC. We do not study the massless field subject to NBC, since under these boundary conditions, $\lambda_c = 0$ for $m = 0$, as pointed out in Ref. [16].

Figs. 5 and 6 display the dynamics of the KGM system in the semiclassical approximation. The corresponding quantities in the external field approximation are given for reference in orange. Notice the different λ scales for DBC and NBC. The system is studied through the energies of the first modes (top panel), the induced charge Q_I (middle panel) and the total electric field $\lambda + Q_I$ at the midpoint $z = 0.5$ (bottom panel). We observe in both cases how considering backreaction avoids the instabilities reflected in $\omega_1 \to 0$ and $Q_I \to -\infty$ as $\lambda \to \lambda_c$ in the external field approximation. We also point the reader to notice how the rapid growth of $|Q_I|$ as λ crosses λ_c keeps $\lambda + Q_I$ below λ_c, which effectively cures the instabilities appearing in the external field approximation. It was surprising to find that $\lambda + Q_I$ decreases instead of stabilizing, which might be a symptom of the breakdown of the semiclassical approximation.

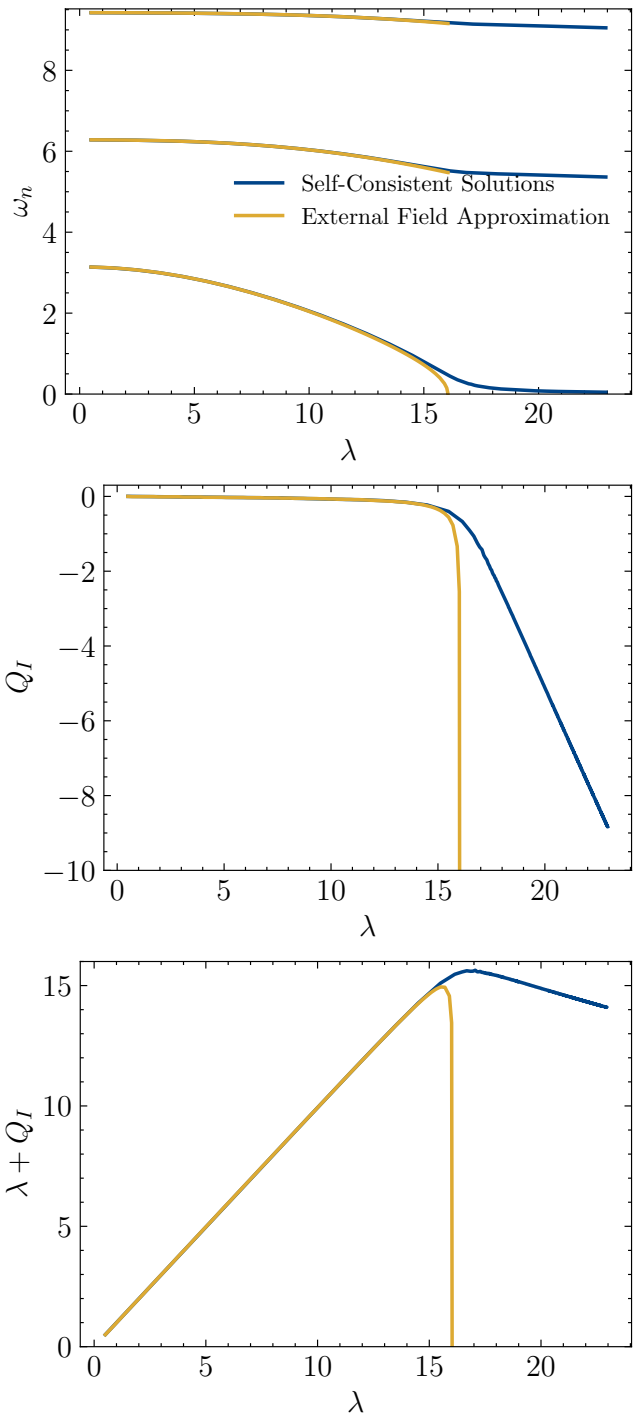

Fig. 5: In blue, the self-consistent solutions to the coupled semiclassical Klein-Gordon-Maxwell system for a massless scalar field subject to DBC. In orange, for reference, the external field approximation is given. Top panel: Energies of the first three modes as functions of λ. Middle panel: Corresponding Q_I for each λ. Bottom panel: Total electric field $\lambda + Q_I$ due to the screening of the vacuum at $z = \frac{1}{2}$.

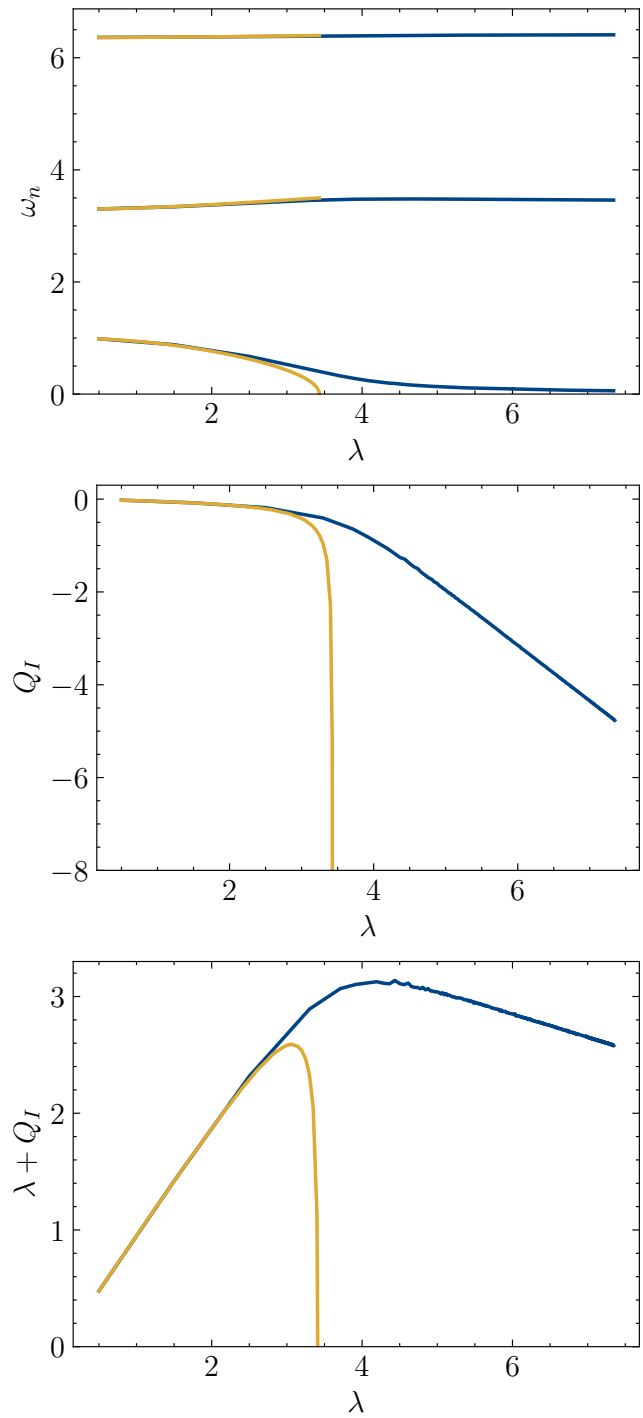

Fig. 6: Following the format in Fig. 5, the results for the unit mass scalar field subject to NBC. Further details in main text.

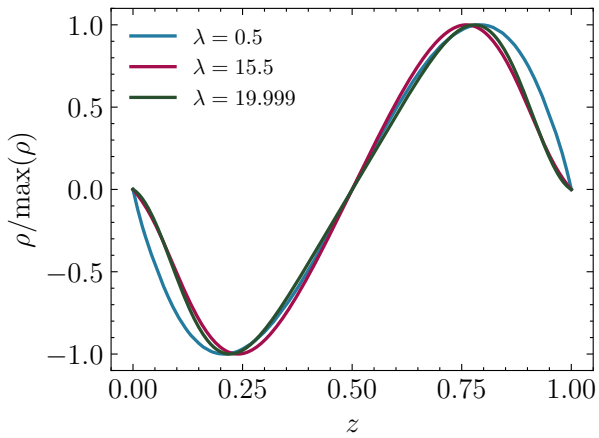

Fig. 7: The self-consistent vacuum polarization for varying λ for the massless field subject to DBC. For visualization purposes, each curve is normalized with respect to their maximum value.

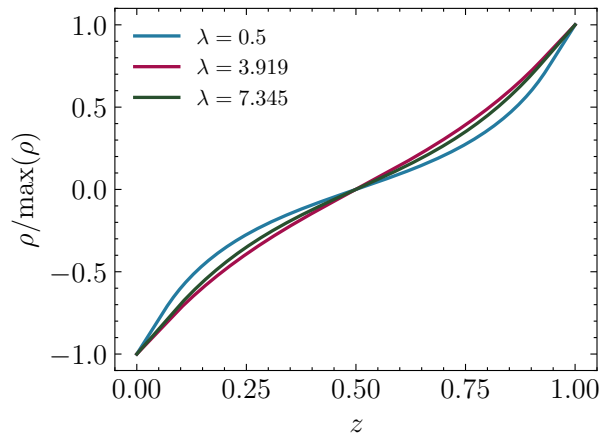

Fig. 8: Same as Fig. 7 for the $m = 1$ field subject to NBC.

One might also notice the almost identical behavior of the self-consistent solutions to the external field approximation in the weak-field regime $\lambda \ll \lambda_c$, which seems to be independent of mass and boundary conditions.

For completeness, the self-consistent vacuum polarizations for different values of λ are displayed in Figs. 7 and 8 corresponding to the results in Figs. 5 and 6, respectively.

5.1 Mode sum formula

We compare the results obtained in this work with those of [1]. Fig. 9 displays Q_I in the iterative procedure as a function of λ for two different prescriptions for vacuum polarization: in blue, the correct prescription ρ used throughout this paper, and in orange ρ_1^{MS} as discussed in Sec. 3.3. The two prescriptions exhibit markedly different behavior in the weak-field regime, where the truncated mode sum yields a noticeably stronger vacuum polarization. In the strong-field regime, however, their behavior is identical. This can be understood as a consequence of the low-energy modes being the most sensitive to the background field, and thus the ones that contribute most to vacuum polarization.

6 Discussion and outlook

Ambjørn & Wolfram [1] studied the effect of backreaction of a charged quantum scalar field due to opposite charges placed on the boundaries of a finite 1-dimensional region. Instabilities appearing in the external field approximation are studied, along with their resolution through vacuum polarization via an iterative procedure. It was pointed out later that their prescription ρ^{MS}, based on a mode sum formula, leads to wrong results [13]. The correct prescription ρ introduced in [16] finds discrepancies of approximately an order of magnitude with ρ^{MS}, in the external field approximation. In this paper, we revisited the semiclassical KGM equations studied by Ambjørn *et al.*, with the correct prescription for ρ under varying boundary conditions and masses of the scalar field.

Our primary finding is that the backreaction through ρ also avoids the instabilities that appear in the external field approximation. However, we observe that beyond the critical field, the total electric field does not stabilize at a fixed value; instead, it recedes. This may indicate a breakdown of the semiclassical approximation, as the quantum fluctuations of the electric field are neglected.

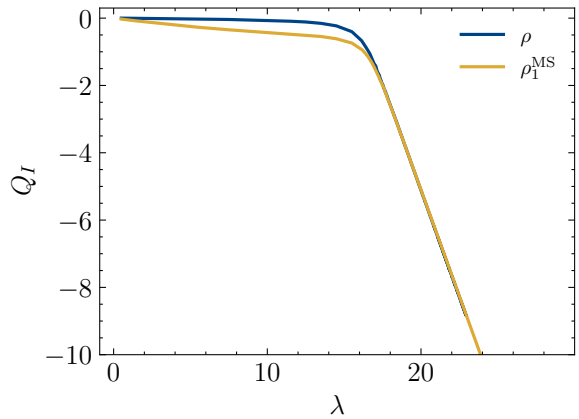

Fig. 9: Q_I of the self-consistent solutions for a massless field subject to DBC resulting from the two prescriptions of vacuum polarization described through the article.

Interestingly, when we compare the results using the correct prescription with those obtained with the older method, their strong-field behaviors are found to be indistinguishable. This is attributed to the low-energy modes being the most sensitive to the background field, which are precisely the modes that dominate the onset of runaway solutions.

Future work could focus on benchmarking the semiclassical approximation with scalar QED. This is a key comparison for delineating the regime of validity of the semiclassical approach.

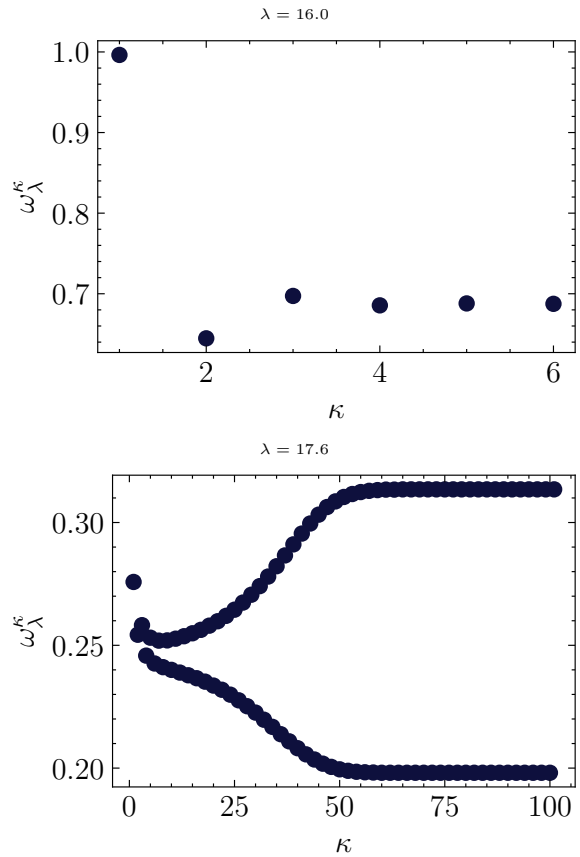

Fig. 10: ω_1 of the massless scalar field with DBC for different values of λ as a function of the iteration κ.

A Convergence of the self-consistent fields

The iterative procedure either converges, oscillates, or breaks down.

The convergence of the iterative procedure described in Sec. 4 is measured through the sequence $\{\omega_{1,\lambda}^\kappa\}_{\kappa>0}$. Examples of a converging sequence and an oscillating (non-convergent) sequence are displayed in the top and bottom panels of Fig. 10, respectively. These cases can be understood from the perspective of fixed-point theory, where we explicitly see whether the update function $f_{\lambda,c}$ is a contraction (top panel) or not (bottom panel). We conjecture that one can always find a small enough c such that the map $f_{\lambda,c}$ is contracting.

The iterative procedure breaks down when complex ω appear. This happens if the candidate $\lambda+\Delta\lambda$ in the numerical continuation method is too far away from λ: the screening due to ρ_λ is too weak compared to $A^1_{0,\lambda+\Delta\lambda}$, and the system effectively behaves as in the external field approximation with $\lambda > \lambda_c$. The step $\Delta\lambda$ is reduced and the iterative procedure is tried again.

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